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Consciousness, Quantum Mechanics, and the Limits of Scientific Objectivism

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Consciousness and quantum mechanics are among the most puzzling phenomena studied in the sciences. Some scholars suggest they are related, though others think this claim commits a "minimization of mystery" fallacy. The aim of this programmatic paper is to draw attention to a less widely discussed parallel between consciousness and quantum mechanics: both challenge the classical objectivist worldview of science. Under certain assumptions, they are each in tension with a package of metaphysical theses – "non-relationalism", "non-fragmentation", and "one world" – that jointly make up that worldview. This points to three distinct non-objectivist responses: the "relationalist", "fragmentalist", and "many-subjective-worlds" ones. We will map out their pros and cons.

1. Introduction

Consciousness and quantum mechanics are both thought to raise significant challenges for science. Consciousness has been described as being "at once the most familiar thing in the world and the most mysterious" (Chalmers 1996, p. 3). Despite progress in neuroscience, we still lack an accepted explanation of how it fits into the physical world. Quantum mechanics makes some of the best-confirmed predictions we find anywhere in science, for instance predicting with extraordinary accuracy how electrons behave in magnetic fields, and it supports technologies from lasers to MRI scanners, and yet there is nothing even remotely resembling a consensus concerning its interpretation. Both fields are riddled with paradoxes and unresolved problems: the "hard problem of consciousness" and the mind-body problem more broadly, the puzzle of Schrödinger's

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cat, the Einstein-Podolsky-Rosen paradox, and various "no-go" results, to name just a few. In fact, some scholars have suggested that consciousness and quantum mechanics are connected (see Gao 2022). According to Penrose (1994), for example, consciousness cannot result from ordinary algorithmic processes but may depend on non-computable processes, possibly involving resources from quantum mechanics. Hameroff and Penrose's proposal is that consciousness stems from quantum effects in tiny microtubules in the brain (2014). Chalmers, on the other hand, attributes the tendency to connect these two fields to the temptation to invoke a "Law of Minimization of Mystery": "consciousness is mysterious and quantum mechanics is mysterious, so maybe the two mysteries have a common source" (Chalmers 1995, section 5). But even Chalmers acknowledges possible connections. Jointly with McQueen, he has recently revived the idea that the collapse of the wave function is triggered by consciousness, an idea that goes back to John von Neumann and Eugene Wigner but which has long been dismissed (Chalmers and McQueen 2022). He further notes that the temptation to see connections between the two fields "is magnified by the fact that the problems in quantum mechanics seem to be deeply tied to the notion of observership, crucially involving the relation between a subject's experience and the rest of the world" (Chalmers 1996, p. 333).

In this paper, we want to draw attention to a less widely discussed parallel between consciousness and quantum mechanics: both put pressure on the classical "objectivist" worldview that is commonly associated with science since the Enlightenment, and they do so in surprisingly similar ways. Our claim is this:

A challenge for objectivism: Consciousness and quantum mechanics, at least under certain assumptions, are each in tension with a package of metaphysical theses – "non-relationalism", "non-fragmentation", and "one world" – that jointly make up the objectivist worldview of mainstream science.

We will consider the argument for this claim in each of the two domains and look at how we may need to revise the objectivist worldview in response. Specifically, we will provide a roadmap of three distinct kinds of non-objectivist approaches: "relationalist", "fragmentalist", and "many-subjective-worlds" approaches.

Our analysis brings together insights from some recent "no-go" results concerning each domain: a "quadrilemma for theories of consciousness" (List 2025) and a "heptalemma for quantum mechanics" (DeBrota and List 2026). These build on earlier results in the relevant fields, especially Fine's (2005) analysis of first-personal facts and Bell's theorem in quantum mechanics (Bell 1964), and so our analysis is indebted to those earlier results too and to the bodies of work they have inspired. Importantly, our focus will be on the big picture and on programmatic issues raised by the challenge for

objectivism. As it is often said, the devil is in the detail, and there will be many details to which we cannot do justice here.

Scholars in the phenomenological tradition have also emphasized the limitations of the objectivist worldview and have discussed problems like the ones discussed here (for recent works, see, e.g., French 2023 and the volumes edited by Berghofer and Wiltsche 2020, 2023). If we arrive in a similar place coming from a different angle, then what we are arguing should be seen not so much as an attempt to reinvent the wheel but rather as independent corroboration of broadly similar ideas. On the other hand, the paper should also be of interest to those who wish to resist the non-objectivist conclusion, because it may help to clarify which premises one must reject in order to do so.

To avoid any misinterpretation, we should stress that we are not suggesting that consciousness is the key to solving the problems of quantum mechanics, or that quantum mechanics is the key to solving the problems of consciousness. What we are suggesting is something more modest: there is a *structural* parallel between the problems in the two areas, and making sense of each set of problems may require a particular kind of departure from the classical "objectivist" worldview of science. If there is indeed such a parallel, this might lend support to a certain kind of non-objectivist approach to science. It is this parallel and any resulting lessons that we want to explore.

2. The hard problem of consciousness

Philosophers have long grappled with what David Chalmers has called the "hard problem of consciousness" (1995, 1996).[1] While science is good at explaining many natural phenomena, from the trajectories of the planets to the functioning of biological organisms, science has struggled to explain how consciousness fits into the world. The problem is not so much to explain why the capacity for cognition and information processing has evolved in animals and humans. That seems perfectly explicable. It will have been evolutionarily advantageous for our ancestors to have that capacity: cognition and information processing are likely to increase an organism's chances of survival and procreation in a complex environment. The problem is to explain why this is accompanied by subjective experiences, the sorts of experiences that we each have from a first-person perspective.

There is something it is like to be a conscious subject, for that subject, as Nagel (1974) famously puts it. Why is this so? Why do we each have a subjective perspective that we consciously experience, over and above the apparently objective physical activity in our brains? Why don't cognition and information processing take place "in the dark",

[1] Our exposition of the challenge draws on some of Christian List's previous works, including List (2023, 2025) and especially List (2025a), from which we have borrowed some wording.

just as a computer engages in information processing without presumably experiencing anything? It is generally thought that the survival value of cognition and information processing can be explained by reference to the *objective* manifestations of the relevant processes and does not depend on the subjective experiences accompanying them: what it feels like to undergo those processes.

The much-discussed thought experiment of a philosophical "zombie" helps to illustrate these points (Chalmers 1996, Kirk 2023). A philosophical "zombie" is a hypothetical creature that is objectively indistinguishable from each of us but that subjectively experiences nothing. It outwardly behaves in the same way each of us does. It produces the same verbal responses we do. Its brain activity, according to EEG and fMRI scans, is the same as ours. However, in this thought experiment, all of this physical activity takes place without an inner stream of first-personal consciousness. Such a hypothetical zombie would produce the same objective responses to its environment as any one of us does, and thus it would survive just as well. If, as Chalmers and others have argued, this thought experiment is logically coherent (notwithstanding its far-fetchedness), it helps us see that the first-person experiences that we each have – i.e., what it *subjectively feels like* to undergo the relevant cognitive processes – is an unexplained "add-on": something that cannot be explained just by pointing to the objective manifestations of cognition and information processing. Expressed slightly differently, conscious experience is first-personal and subjective, while ordinary science is third-personal and objective. It seems that a third-personal and objective explanation, of the sort science normally offers, cannot explain a first-personal and subjective *explanandum*.

One response to this problem is to argue that, to accommodate consciousness in a philosophical or scientific worldview, we must assume that consciousness is somehow fundamental rather than derived. Chalmers (1995, 1996) has prominently defended one version of this response. Incorporating consciousness into a scientific worldview, he suggests, would require a theoretical innovation similar to the one made by James Clerk Maxwell and others when they first postulated the existence of electromagnetic fields. What Maxwell noted was that the inventory of properties that exist in the physical world was richer than recognized in previous physical theories such Newton's classical mechanics. There are not only the Newtonian properties related to gravity and motion, but also electromagnetic properties that were missing from Newton's theory. Similarly, Chalmers claims, the world contains not only the generally recognized physical properties, but also distinct "phenomenal" properties, which are responsible for conscious experiences. According to Chalmers, it is the existence of those phenomenal properties that accounts for the difference between the actual world, in which there is consciousness, and the

"zombie world" of the thought experiment, in which everything is physically the same but consciousness is absent. In effect, Chalmers proposes a modern, scientific version of dualism, thereby updating the sort of traditional dualist view associated with Descartes.

Chalmers's work has reframed the debate about consciousness around the question of what kinds of properties must exist in the world to account for consciousness along with everything else. The basic philosophical disagreement, in this framing, is that between physicalists and non-physicalists. Physicalists think that it is unnecessary to postulate any non-physical properties over and above the ordinary physical ones: once all physical properties are in place as they are, consciousness is an automatic byproduct, albeit a highly complex one. Non-physicalists, like Chalmers, object that this worldview fails to do justice to the intuition that the thought experiment of a philosophical zombie is coherent (albeit purely hypothetical) and that there remains a gap between what could be explained by reference to physical properties alone and what must be explained in order to account for consciousness. Consequently, non-physicalists such as Chalmers think we must postulate phenomenal properties as additional fundamental building blocks of the world or give an enriched account of the nature of the physical.[2]

These scholarly disagreements, however, obscure one important point of agreement (as noted in List 2023): virtually all sides in the debate share one key assumption, namely that we can think of reality as being exhausted by a single unified and coherent world, which science and philosophy seek to describe and explain and which is "populated" by all the facts to be explained. Call this the "objectivist" assumption; we will make it more precise later. It seems that almost everyone in the analytic tradition, from the most hardline physicalist at one end of the spectrum to the staunchest dualist at the other, accepts this assumption or at least tacitly takes it for granted. The main disagreements merely concern the precise ontological inventory of that "objective world": what facts or properties populate it; whether it is it exhausted by physical facts and properties alone; or whether it contains something over and above the physical, such as special phenomenal properties or properties that somehow possess both intrinsic and extrinsic aspects. The heated nature of this debate makes it easy to overlook the shared objectivist assumption that sits in the background.

Why does this matter? It matters because all of the mainstream scientific and philosophical theories of consciousness, including some of the most popular non-physicalist theories such as Chalmers's, give us a fundamentally third-personal or impersonal picture of the world. They attempt, in effect, to describe and explain "the world" as it

[2] A popular strategy is to postulate that there is one kind of fundamental property, but that it has not only ordinary physical aspects – so-called functional or extrinsic ones – but also phenomenal or intrinsic aspects, which underpin consciousness. For an overview of such "monist" theories, see Mørch (2024).

would be seen by an omniscient, or highly informed, outside observer: someone who is taking what Nagel (1986) has called "the view from nowhere".

The attempt to abstract away from any particular perspective is, of course, a central feature of science as we know it. The goal is to identify general patterns and regularities: ones that are invariant under shifts in perspective, i.e., which remain stable as we move from one perspective to another, and which thereby qualify as "objective". Newton's laws of motion are examples. To identify such patterns and regularities, we must abstract away from the particularities of any specific objects, events, and perspectives. We must aim to arrive at a non-perspectival worldview – the sort of worldview for which Nagel coined the term "the view from nowhere". The history of science illustrates this process of abstraction, with the transition from an anthropocentric to a geocentric to a heliocentric and ultimately to a more universal view of the world, and by and large this has been a success story.

Nevertheless, there appears to be at least one phenomenon that resists this kind of objectivization or "de-perspectivization", and that is the phenomenon of consciousness itself. Later we will consider whether some quantum-mechanical phenomena fall into a similar category. In the case of consciousness, the defining feature is subjective experience: there is something it is like to be a conscious subject, for that subject, as Nagel notes. This puts subjectivity at the heart of consciousness. The phenomenologists in the tradition of Edmund Husserl (e.g., 1970) have been well aware of this point (Zahavi 2025) and have very much resisted the idea of a "view from nowhere". For them, the first-personal character of experience is central and what we call "the objective facts" are an emergent abstraction that corresponds broadly to intersubjective agreement. The core of my, say Christian's, consciousness lies in the fact that I find myself in a world in which there are first-personal facts. I am conscious, I have certain experiences, I am in a particular perceptual state, and so on. First-personal facts are irreducibly subjective. They are "centred" around my perspective as an experiencing subject, and unlike objective facts, they are not invariant under shifts in perspective. If we shift from one perspective to another, the first-personal facts change. The "objective" facts are broadly those that survive such shifts.

If this is right, then the correct move to accommodate consciousness in our worldview may not be to postulate a new class of properties – phenomenal properties – within the objective world as ordinarily understood, but rather to postulate irreducibly first-personal facts as a feature of reality: facts that have an inbuilt conscious perspective, facts which are "centred" around a subjective perspective.[3] On this picture, "[r]eality is

[3]Nagel already recognized that just enriching the postulated inventory of the objective world, such

not exhausted by the 'objective' or impersonal facts but also includes facts that reflect a first-person point of view", as Kit Fine puts it (2005, p. 311). Fine calls this thesis "first-personal realism", albeit without endorsing it.

A simple example of a first-personal fact, which holds where I, Christian – currently the narrator – stand, is the fact that I am seeing a beige computer screen as I am writing this right now. This must not be confused with the third-personal fact that Christian is seeing a beige computer screen at a specific point in time. The latter is a third-personalized version of the given first-personal fact, as it continues to hold even if we step away from Christian's particular first-person perspective. The third-personal fact is true for John and everyone else as much as it is true for Christian. It is true, from where you stand, that Christian is seeing a beige computer screen at this particular time. However, assuming that you are not looking at a beige computer screen, "I am seeing a beige computer screen right now" is not a first-personal fact for you.

Postulating first-personal facts is still very heterodox in both science and analytic philosophy. According to the mainstream view, the first-person/third-person distinction cannot be drawn at the level of facts at all but is just a linguistic or cognitive distinction. There are first-personal *modes of representation* of some facts, but the facts themselves are always third-personal or impersonal; facts themselves never come with an in-built perspective. According to the orthodox view, there is no distinct first-personal fact such as "I am seeing a beige computer screen right now". The sentence in quotation marks is just a first-personal way of expressing the impersonal fact that Christian is seeing a beige computer screen at this particular point in time.

Some philosophers, however, think that our picture of reality is incomplete unless we accept that there are genuine first-personal facts, and not just their third-personalized counterparts. We miss out on an important aspect of reality if we don't recognize that there are irreducibly first-personal facts. For philosophers in the phenomenological tradition, as already noted, the first-person perspective is central (see Zahavi 2025). Once we recognize that the first-person perspective is our window to the world and that all experience is first-personally given, it becomes natural to think that any enquiry must begin with first-person facts and can arrive at third-person facts only subsequently, by abstracting away from the particularities of any first-person perspective (for instance, by asking which facts are invariant under changes in the first-person perspective). This was approximately Husserl's picture. French (2023, p. 186) characterizes "the subjective

as along dualist lines, is insufficient. In his words (quoted and discussed in Ratcliffe 2002, p. 356): "The broader issue between personal and impersonal, or subjective and objective, arises also for a dualist theory of mind. The question of how one can include in the objective world a mental substance having subjective properties is as acute as the question how a physical substance can have subjective properties."

stance" as one of "the core features of phenomenology". Among analytically oriented philosophers Lynn Rudder Baker (2013) is one of the few who have defended the existence of first-person facts. She argues that because the scientific worldview "takes the world to be impersonal", it cannot properly "accommodate me", as a conscious subject (2013, xiii–xiv). To accommodate subjects and their perspectives, she suggests, we must accept that "there are irreducible first-person facts" (ibid.). Nagel, similarly, notes that a purely third-personal view of the world leaves out "the fact that I am Thomas Nagel" (1965, p. 355; on parallels between Nagel and Husserl, see Ratcliffe 2002). Nagel further writes:

> "The reductionist program that dominates current work in the philosophy of mind is completely misguided, because it is based on the groundless assumption that a particular conception of objective reality is exhaustive of what there is." (Nagel 1986, p. 16)

Indeed, in a purely third-personal worldview, there is no room for the fact that I am who I am, or that I occupy a particular perspective on the world. There is only room for the fact that Christian occupies a particular perspective, or that John occupies a particular perspective, but as already noted, that's not the same as the fact that *I* occupy such-and-such perspective.

More recently, however, there has been a growing interest in realism about first-personal facts. Martin Lipman (2023), for example, asks: "What shape must reality take for there to be a place for consciousness in it?" (p. 530). In answer, he writes: "for the world to harbor consciousness, reality must include subjects that are metaphysical standpoints and be such that some of the facts obtain only from the standpoint of certain subjects and not others" (pp. 530–531). He sums up the view by saying that "reality is constituted by subjective facts" (p. 531). Others who have explored similar views include Hellie (2013), Merlo (2016), Conitzer (2020), one of us (List 2023), and Builes (2024).

Yet, once we postulate first-personal facts, we face a significant theoretical challenge. We must reconcile the existence of first-personal facts with the objectivist view of reality that we traditionally find in science and philosophy or alternatively revise that view. It is this challenge that we now want to discuss.

3. The challenge for objectivism

Let us start with the following thesis:

First-personal realism: Reality includes, for each conscious subject, some first-personal facts.

We want to examine how this relates to the standard objectivist worldview of science and philosophy. That worldview, we propose, consists of three theses about the nature of reality, which we call "one world", "non-fragmentation", and "non-relationalism" (following the terminology in List 2025 and DeBrota and List 2026). The three theses are so widely accepted that they are seldom explicitly stated. The first expresses the core idea that there is one objective world.

One world: Reality is exhausted by one world (which we may think of as an objective world), not multiple worlds (which we may think of as subjective ones).

Although usually left implicit, this thesis is explicitly discussed, for example, in Strawson (1966, part 2, ch. 3, section 8), who, in turn, refers to its Kantian origins. Commenting on those origins, Moore (2016) quotes Kant's statement that "all appearances lie, and must lie, in *one* nature" (Kant, transl. by Smith 1929, A216/B263) and interprets it as the thesis that "the Cosmos constitutes the whole of empirical reality", where "the Cosmos" is defined as "the four-dimensional realm that we inhabit" (Moore 2016, p. 934).[4]

The second thesis adds an important qualification about the nature of any such world.

Non-fragmentation: Any world (actual or possible) is a coherent collection of facts. Equivalently, the total collection of facts that obtain at any given world can be coherently coinstantiated.

This idea that any possible world must be coherent is very much in the spirit of the Kantian picture mentioned above. The relevant notion of a world is nicely articulated in the opening sentence of Wittgenstein's famous *Tractatus*: "The world is everything that is the case", which he further clarifies by adding that "[t]he world is the totality of facts, not of things" (1922, 1 and 1.1). The coherence of any world, as required by non-fragmentation, is essential to that notion. A world that is not a coherent totality of facts would not be a possible world as conventionally understood, let alone a candidate for being the actual world. Fine (2005) uses the name "coherence" for a version of the non-fragmentation thesis.

The third thesis adds another important qualification:

Non-relationalism: Any fact within the collection of facts that constitute reality is of the absolute form "such and such is the case", not of the relative form "such and such is the case, relative to such and such".

[4]Moore points out that "Kant himself allows for the possibility that the Cosmos does not exhaust reality in a broader sense of 'reality' "(ibid.). We will consider that possibility later in the present paper.

This thesis can also be motivated by reference to Wittgenstein's understanding of facts. Recall that, for Wittgenstein, a fact is *something that is the case*. It is something that is the case *simpliciter*, not something that is the case only relative to something else. Examples of facts are the following: there is life on Earth; carbon dioxide contains one carbon atom and two oxygen atoms; London is a city; $1 + 1 = 2$. Each of these facts holds *simpliciter*, not relative to something else. There is life on Earth, full stop. This is not merely the case relative to something else. Carbon dioxide has the stated composition, full stop, not just relative to something else. To understand the difference between a fact in this absolute sense and a merely relative fact, contrast "carbon dioxide contains two oxygen atoms" with "carbon dioxide contains more oxygen atoms". The first expresses a fact in the absolute sense. The second invites the question: relative to what? Relative to carbon monoxide, carbon dioxide contains more oxygen atoms. But relative to carbon trioxide or sulfur trioxide, it does not. Without any relativization parameter, the sentence "carbon dioxide contains more oxygen atoms" does not express a fact in the ordinary sense. Non-relationalism asserts that the facts making up reality are of the absolute form "such and such is the case", not of the relative form "such and such is the case relative to such and such". Fine (2005) call this thesis "absolutism".

Let us use the label "objectivism" for the conjunction of the three theses just stated: one world, non-fragmentation, and non-relationalism. Objectivism thus defined is an ontological rather than epistemological thesis. That is to say, it is a thesis about what reality is like, not a thesis about what we can know about reality and how we can do so. If objectivism is true, then reality consists of one world, which we can think of as a coherent and exhaustive collection of facts that are instantiated at that world and where those facts have an absolute rather than relative form. What we can come to know about reality, and how we can do so, remains a separate question, which is epistemological and not our concern here, although it is an important question in its own right.

We now have all the background in place to see the tension between first-personal realism and objectivism. The tension arises under the reasonable assumption of non-solipsism:

Non-solipsism: More than one conscious subject is real; in particular, I am not the only conscious subject.

Our claim (originally defended in List 2025) is that first-personal realism, non-solipsism, and objectivism are jointly inconsistent.[5] To see this, suppose we accept

[5]In contrast to the exposition in List (2025), we are here combining three theses (one world, non-fragmentation, and non-relationalism) into a single thesis (objectivism). Further, in List (2025), non-relationalism was not stated as a separate thesis but was treated as a presupposition of first-personal

both first-personal realism and non-solipsism. Then more than one conscious subject is real. For each conscious subject, there are first-personal facts. Furthermore, given non-relationalism, these first-personal facts obtain *simpliciter*, not relative to anything else. For example, the fact that I – say Christian – am Christian is the fact that

(i) I am Christian, full stop.

It is not merely the fact that

(ii) I am Christian, relative to Christian's perspective,

or the fact that

(iii) relative to Christian's perspective, it is true to say "I am Christian".

Of course, what is expressed by sentences (ii) and (iii) is true, but it is not a genuinely first-personal fact. Even from where John and everyone else stands, it is true that "relative to Christian's perspective, it is true to say 'I am Christian'". Just as the fact that Christian is seeing a beige computer screen at a specific time is a third-personal fact, so the facts expressed with the help of the "relative to"-clauses, i.e., (ii) and (iii), are not genuinely first-personal either. Only "I am Christian" expresses a first-personal fact; "I am Christian, relative to Christian's perspective" or "relative to Christian's perspective, it is true to say 'I am Christian'" do not.

With this qualification in place, it becomes evident that different subjects' first-personal facts are mutually incompatible. As an illustration, consider the first-personal fact, which holds from where I, Christian, stand, that I am seeing a beige computer screen in front of me right now. Suppose that you are not seeing a beige computer screen in front of you right now; you are seeing something else. Then, from where you stand, the following is true: "I am not seeing a beige computer screen in front of me right now". The two first-personal facts ("I am seeing a beige computer screen in front of me right now", and "I am not seeing a beige computer screen in front of me right now") cannot be co-instantiated as first-personal facts. Only their third-personalized counterparts are compatible: Christian is seeing a beige computer screen in front of him right now; Christian's interlocutor is not seeing a beige computer screen in front of them right now. Or alternatively, their relativized versions are compatible: "relative to Christian's perspective, it is true to say 'I am seeing a beige computer screen in front of me right now'" and "relative to the perspective of Christian's interlocutor, it is true to say 'I am not seeing a beige computer screen in front of me right now'". But these are

realism; see especially section III of that paper. We will come back to this point later.

not first-personal facts; they are third-personalized or merely relational counterparts of the given first-personal facts. So, we have established the following:

Lemma: Given non-relationalism, the first-personal facts of different conscious subjects are not all mutually compatible.

Since first-personal realism commits us to realism about first-personal facts, not merely to realism about their third-personalized or relational counterparts (noting again that we have assumed non-relationalism), we get the conflict with the claim that reality is exhausted by one coherent world. The total collection of facts constituting reality, which includes different subjects' first-personal facts, is not coherent. Therefore, we cannot simultaneously accept one world and non-fragmentation. If reality is exhausted by a single world, this will be incoherent. And if worlds are each supposed to be internally coherent, then reality cannot be exhausted by a single such world.

Summarizing these observations, we get a no-go result, which is a variant of the result in List (2025). A precursor of the result can be found in Fine (2005).[6]

No-go result for consciousness: First-personal realism, non-solipsism, and objectivism are jointly inconsistent. Any two of the three theses are jointly consistent.

To see that any two of the three theses are jointly consistent, it suffices to consider the possibilities of satisfying any pair of them. The least compelling possibility is arguably the denial of non-solipsism. If we are prepared to suppose that reality admits only a single conscious subject – myself – then the issue of how to reconcile different subjects' incompatible first-personal facts no longer arises. We can then postulate a single coherent world that accommodates all of my (the only subject's) first-personal facts along with all facts that we would conventionally classify as third-personal or impersonal, in a way that that is formally consistent with how we have defined objectivism. Caspar Hare (2007) has defended such a solipsist view, which he calls "egocentric presentism". While the view is consistent (and it's important to acknowledge it as a possible position), however, this is unlikely to be a route many would want to take. As Fine notes, it seems "metaphysically preposterous that, of all the people there are, I am somehow privileged – that my standpoint is the standpoint of reality and that no one else can properly be regarded as a source of first-personal facts" (2005, p. 285).

[6]Fine notes that first-personal realism conflicts with three other theses that are similar to the ones discussed here: "first-personal neutrality", which roughly corresponds to non-solipsism; "absolutism", which corresponds to non-relationalism; and "coherence", which corresponds to non-fragmentation or perhaps the conjunction of non-fragmentation and one world. Fine does not explicitly state the thesis of one world but seems to presuppose that reality is exhausted by a single world.

The denial of first-personal realism, by contrast, is very popular. It can be found in most mainstream theories of consciousness in analytic philosophy, most notably in physicalist theories but also in Chalmers's dualistic alternative, as already discussed in the previous section. Scientific theories that seek to identify consciousness with certain neuronal properties or patterns of brain activity also fall into this camp, for instance those that say "consciousness *is* synchronized brain activity in a certain frequency range". Those theories embrace standard scientific objectivism, and they are non-solipsistic. For reasons of consistency, they must then reject first-personal realism; and indeed, they do. Interestingly, with the exception of so-called illusionist theories (Frankish 2016), which explicitly regard consciousness as an illusion, the mainstream theories still claim that they consider consciousness to be real.[7] Nevertheless, by rejecting first-personal realism they are committed to a certain form of anti-realism about first-personal facts, whether or not they admit this. Anyone who is persuaded by the centrality of subjectivity to consciousness may therefore be inclined to conclude that those theories fail to accommodate consciousness properly. This leaves the denial of objectivism as a serious possibility.

4. Non-objectivist approaches to consciousness

Given that objectivism is the conjunction of three theses, any one of them can in principle be relaxed while upholding the other two. Perhaps the least radical departure from objectivism, in some respects, is to uphold one world and non-fragmentation and to argue that first-personal facts are only relative rather than absolute facts. We would thus relax non-relationalism, but this relaxation would be restricted to first-personal facts. The mutual incompatibility of different subjects' first-personal facts would then be avoided insofar as there is no conflict between it being true relative to Christian's perspective that "I am seeing a beige computer screen right now" and this not being true relative to someone else's perspective, such as John's. But while this is a coherent position, it is unclear whether it does justice to the nature of first-personal facts. Given the proposed relaxation of non-relationalism, first-personal facts would only be facts of a relativized sort, such as "relative to Christian's perspective, it is true to say 'I am having Christian's experiences'", and those relativized facts are arguably not genuinely first-personal. Even the total collection of such relativized facts would leave out the fact that *I am having Christian's experiences rather than anyone else's*. To illustrate, consider the collection of facts:

[7] Illusionism raises the question of whether there can be an illusion without a conscious subject for whom it is an illusion. Illusionists must, in effect, reject the commonsense reading of this question.

- relative to Christian's perspective, it is true to say "I am having Christian's experiences";
- relative to John's perspective, it is true to say "I am having John's experiences";
- relative to Alice's perspective, it is true to say "I am having Alice's experiences";
- and so on.

This collection of facts leaves open which experiences *I* have. It does not settle the question of whether I am Christian, or John, or Alice, or someone else. This is because the relativized facts are still inventorized by an Olympian observer adopting a view from nowhere; they do not pick out any perspective at all.[8] Reconstruing first-personal facts as relational facts would thus amount to a denial of first-personal realism in the originally intended sense. Moreover, if we went along the relationalist route, we would still have to answer the question of what exactly the relativization parameter is. Presumably, it would be a "subject" or a "subjective perspective", but then we would need to explain how subjects or subjective perspectives fit into the world. Interestingly, this is where Fine (2005) sees the greatest difficulties with the relationalist position. He writes:

> "If we opt for the relativist position, then we must take each subjective reality to be given relative to a metaphysical subject or self. But reality itself contains no metaphysical self. We therefore arrive at the conception of the pure metaphysical self – one that stands outside the world and yet is that by which the world (or the subjective world) is given. One can see why philosophers might have been attracted to such a position, given that they wished to give proper recognition to the multiplicity of different subjective viewpoints that, in themselves, were without a point of view. But the position is barely intelligible; and the mystery of the pure metaphysical self no longer arises once we opt for the fragmentalist [as distinct from the relativist] position." (p. 314)

Indeed, most first-personal realists in the small scholarly literature on the subject do not take the relationalist route. Setting aside solipsism, the most popular route appears to

[8] Another way of expressing the same point is to say that the relativized facts, unlike genuinely first-personal facts, do not settle what Hellie (2013) has called the "vertiginous question": why am I having *my* experiences rather than anyone else's? Now, perhaps Hellie's question has no satisfactory answer. But at least we might want to insist that it is a fact *that* I am having my experiences rather than anyone else's, even if this is a brute fact for which one can give no further explanation. The relationalist approach would not be able to recognize such a brute fact. According to it, there would only be the relative fact that relative to being Christian, it's true to say "I am having Christian's experiences", but that's not the fact we are after if we take the vertiginous question seriously.

be the relaxation of non-fragmentation, as already anticipated in Fine's quoted passage. This would allow us to uphold the view that reality is exhausted by one world and that all facts making up that world, including all first-personal facts, are non-relational. We would have to concede, however, that the world is incoherent. This seems metaphysically costly, given that the notion of coherent worlds is so central to logic, philosophy, and science. Yet, the position does do justice to the idea that I am having my own experiences *simpliciter*, and not just in some relativized sense. And indeed, if a first-personal realist rejects both solipsism and relationalism and wishes to uphold the standard idea of a single world, then they must embrace a fragmentalist worldview. This is also the conclusion to which Fine (2005) is led by his analysis, and so is Lipman (2023). To make fragmentalism more palatable, one might say that although reality as a whole is a globally incoherent collection of facts, it has locally coherent subcollections: coherent fragments. Merlo (2016, p. 324) characterizes the relevant metaphysical picture along similar lines: "one could take all points of view to be on a par vis-a-vis truth simpliciter by treating them as different 'fragments' of an overall incoherent totality of facts".

A third route is to accept that there is not just a single third-personal or objective world, but many different first-personally centered or subjective worlds, corresponding to different conscious subjects. Each of these subjective worlds can then still be internally coherent and individually made up of non-relational facts, so that we can uphold non-fragmentation and non-relationalism. This route breaks with the traditionally assumed identity "reality = the world". According to it, reality consists of a collection of subjective worlds, not a single objective world. This would be a many-worlds theory of consciousness, or a many-first-personally-centred-worlds theory (as sketched in List 2023). The resulting worldview is obviously very heterodox, and metaphysically costly in other respects. Arguing for it would require a balancing of considerations. In particular, if first-personal realism, non-solipsism, non-relationalism, and non-fragmentation seem too implausible to abandon, then the many-subjective-worlds route becomes a plausible contender. It is worth noting that a few philosophers have previously considered the possibility of giving up the assumption of one world. Strawson recognizes the question "[w]hy only *one* unified objective world" as a genuine one and acknowledges some rival possibilities, including "a multiplicity of objective worlds" (1966, p. 151). Goodman (1978, 1984) develops the "pluriworldist" view that there are multiple actual worlds (Declos 2019), and Lewis (1986) famously defends the modal realist thesis that many possible worlds are real, even though only one is actual. (Lewis's account, however, does not refer to subjective or first-personally centred worlds, and so his proposal is still quite distinct from what we are discussing here.) More recent contributions that give

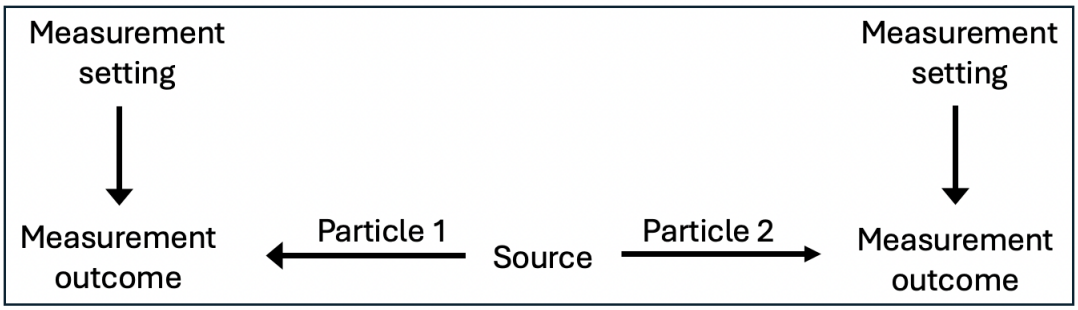
up the idea of one objective world, including in the context of the mind-body problem, are Vacariu's (2005, 2016) account of "epistemologically different worlds", Honderich's (2014) account of "subjective physical worlds", and Gabriel's (2015) argument that "the world", understood as a unified totality of everything, does not exist.

All three non-objectivist routes – the relational route, the fragmentalist route, and the many-subjective-worlds route – are likely to be outside the metaphysical comfort zone of any proponent of the standard objectivist worldview, all the more so because that worldview has served science so well ever since the Englightenment. It is therefore unsurprising that the dominant approach in the analytic philosophy of consciousness would rather reject first-personal realism than entertain a departure from objectivism. And indeed, if consciousness were the only phenomenon for which such a departure might be needed, then the move away from objectivism might seem *ad hoc* and too costly. However, this is where it is instructive to consider the case of quantum mechanics.

5. The EPR paradox and Bell's theorem

To introduce the puzzles of quantum mechanics, it is helpful to begin with the Einstein-Podolsky-Rosen (EPR) paradox (1935). Suppose we set up a device that emits two maximally entangled particles travelling in opposite directions, as illustrated in Figure 1. Once they are very distant from each other, an observer, say Alice, performs a

Figure 1: The setup in the EPR paradox



measurement on the first particle, and another observer, say Bob, performs a measurement on the second. Alice and Bob can each choose between two distinct measurement settings. They can measure either the position or the momentum of their respective particle. So, Alice could choose to measure her particle's position while Bob could choose to measure his particle's momentum, or vice versa, or they could both make the same choice. Now, because the two particles are maximally entangled, if Alice measures position, she can perfectly predict the outcome of a position measurement performed by

Bob; or, if she measures momentum, she can perfectly predict the outcome of a matching measurement on Bob's part too. The same is true, symmetrically, for Bob. Alice's and Bob's measurement outcomes will be perfectly correlated if they choose matching measurement settings. Moreover, since the two particles are far apart at the time of measurement, Alice's measurement should not be able to affect any property of Bob's particle instantaneously and vice versa. Clearly, if the two particles were billiard balls in classical physics with perfectly correlated properties, the situation would be completely unremarkable as each ball is taken to possess position and momentum properties which can be read out by measurement. If Alice were to measure the position or the momentum of one of the two billiard balls, and she knew the other ball had perfectly correlated properties, then of course she would be able to predict the outcome of a matching measurement of the second ball. Einstein, Podolsky, and Rosen assumed that perfect predictability in the absence of any physical influence implies the existence of some element of physical reality corresponding to the predictable quantity. In the case of the maximally entangled particles, then, the perfect predictability of either a position or a momentum measurement of the distant particle, without any physical influence, suggests that the distant particle possesses both position and momentum values, just like perfectly correlated billiard balls.[9] However, variables representing values for these unobserved measurements are not present in the formalism of quantum mechanics. In fact, Heisenberg's famous uncertainty principle, according to which one cannot measure both a particle's position and its momentum with perfect precision, is sometimes interpreted to imply that a particle does not even *possess* these properties unless or until they are measured. Einstein, Podolsky, and Rosen concluded that quantum mechanics must be incomplete. There must be some hidden facts, not yet acknowledged by the standard theory, that determine what the various measurement outcomes would be.

Einstein, Podolsky, and Rosen's analysis leaves open the possibility that the problem might be solved by postulating such hidden facts ("hidden variables"). But a subsequent no-go result, Bell's theorem, shows that this possibility is very limited (Bell 1964, Myrvold, Genovese, and Shimony 2024). In particular, Bell's theorem shows that three widely accepted physical theses are jointly inconsistent with the predictions of quantum mechanics:

Locality: Measurement outcomes in one location are probabilisticially independent of any measurement settings and outcomes in spacelike-separated locations, conditional on any underlying hidden variables.

[9] And this is so, even though Alice would only be able to predict the actual value of one property or the other, depending on which she actually measures.

Measurement independence: Different observers' measurement settings are probabilistically independent of each other and of any property of the system that is being measured.

Realism: Any observer's measurement outcomes correspond to objective facts about the system with well-defined probabilities.

In slightly more technical terms, Bell proved that if all three theses are true of a given system, then the correlations between different observers' measurement outcomes under different possible measurement settings must satisfy certain constraints – the so-called Bell inequalities. But quantum mechanics predicts breaches of those constraints. In essence, quantum-mechanical systems can display stronger correlations than permitted if locality, measurement independence, and realism are true. Therefore, at least one thesis must be false.

Let us briefly review the options. Physics has long emphasized locality as a desirable assumption about physical reality, and locality holds in Einstein's special and general theories of relativity. It is generally considered undesirable to admit any correlations that cannot be accounted for by local transmission mechanisms – any "spooky action at a distance", as Einstein famously put it. Still, some physicists have interpreted Bell's theorem as showing that the physical world is non-local. Maudlin (2014), for instance, argues that the "proper conclusion" is that "any world that displays violations of Bell's inequality for experiments done far from one another must be non-local" (p. 1). Indeed, several prominent interpretations of quantum mechanics give up locality. The de Broglie-Bohm interpretation, for example, depicts quantum-mechanical systems as deterministic systems whose state evolves on the basis of their global particle configuration; this can then account for non-local correlations (Goldstein 2025). Similarly, Ney's wave function realism (Ney 2021) gives up locality, by taking the state of a quantum system to be represented by a point in a higher-dimensional configuration space, distinct from the conventional three-dimensional physical space. From the perspective of our three-dimensional space, we can thus see violations of locality. However, many physicists consider violations of locality problematic. Newton endorsed the idea of locality, even though he didn't use the term. He was famously unhappy about the fact that in his theory gravitational forces have instantaneous long-distance effects (Berkovitz 2016), a shortcoming that Einstein eliminated in his general theory of relativity. In fact, Einstein argued that giving up locality "would make impossible the idea of the existence of (quasi-) closed systems and, thereby, the establishment of empirically testable laws in the sense familiar to us" (quoted and translated in Howard 1985, p. 188).

This leads many to prefer a different response to Bell's theorem. A second option is to

give up measurement independence. If measurement independence does not hold, Alice's and Bob's choices can be correlated even though they are far apart. So-called superdeterministic interpretations of quantum mechanics take this view (see, e.g., Hossenfelder and Palmer 2020). They assume that different observers' choices are correlated due to some underlying hidden variable. The most radical version of this view would say that their measurement choices are completely predetermined. Measurement independence, however, is a deeply entrenched assumption of science. Denying it has been thought to amount to a denial that the experimenters have free will in choosing their measurement settings, and many physicists worry that postulating the required correlations between measurement settings would involve conspiratorial thinking. Maudlin, for instance, argues that the postulated correlations "would require a massive coincidence, on a scale that would undercut the whole scientific method" (2014, p. 22). Similarly, Sen and Valentini (2020, p. 13) argue that "at least as they are currently understood, superdeterministic models are conspiratorial", which leads them "to conclude that superdeterminism, as a possible explanation of the Bell correlations, is scientifically unattractive" (ibid.). A different way of relaxing measurement independence invokes the idea of retrocausality (Price and Wharton 2015, Friederich and Evans 2023). The proposal is roughly the following. While different observers' measurement settings may be correlated with some underlying hidden variable, the direction of causality doesn't go from the hidden variable to the measurement settings but the other way around. So, it is the measurement settings that causally affect the hidden variable, not the hidden variable causally affecting the measurement settings. Postulating a reversed direction of causality, however, is metaphysically costly in its own way, and it is unclear that this move can fully address the worry of conspiratorial thinking.[10]

The limitations of the foregoing responses to Bell's theorem and the depth of the commitment to locality and measurement independence have prompted some physicists to consider relaxing Bell's third thesis, namely realism. Proponents of the so-called Copenhagen interpretation, for example, fall into this camp. But this move faces some objections too. Maudlin (2014) writes: "it has become fashionable to say that another way to avoid Bell's result and retain locality is to abandon *realism*. But such claims never manage to make clear at the same time just what 'realism' is supposed to be and just how Bell's derivation presupposes it" (p. 23). To deny realism, Maudlin suggests, we would have "to find any tacit physical claim [in Bell's argument] that can be somehow denied" (ibid.). As an example, he gives the assumption "that experiments have unique

[10]The worry is often presented as a "fine-tuning worry" (cf. Friederich and Evans 2023). The postulated retrocausal effects must account for the correlations predicted by quantum mechanics but without permitting any superluminal signalling.

outcomes", an assumption denied by the Everett interpretation of quantum mechanics, but one which, he notes, not only Bell but most people make "when considering the significance of Bell's result for the real world" (ibid.). However, he warns, it would be unhelpful to deny realism in a way that boils down to "denying that the standard predictions [of quantum mechanics] are actually correct" (2014, p. 23).

But even if many physicists are reluctant to deny Bell's realism, we should consider this response further, since the other responses to Bell's theorem face serious objections. It is here that we will re-encounter the challenge for objectivism. We will frame this challenge with the help of a recent no-go result that extends Bell's theorem, a "heptalemma for quantum mechanics", developed in DeBrota and List (2026).

6. The challenge for objectivism

We have seen that if we wish to accommodate the predictions of quantum mechanics while upholding locality and measurement independence, we must reject Bell's thesis of realism. To explore what this means for objectivism, we will make use of the insight (from DeBrota and List 2026) that Bell's realism thesis can be decomposed into several more fine-grained subtheses. Specifically, it can be viewed as a conjunction of four subtheses, each of which can be individually relaxed without relaxing the others. The first expresses realism's basic ontic claim and the subsequent three then impose further constraints on it. All four subtheses are so deeply entrenched in mainstream science that they are seldom explicitly articulated. Let us go through them.

The basic ontic thesis expresses the core idea of realism, namely that measurement outcomes are real in the sense that they correspond to some kinds of facts (aspects of reality), perhaps in the weakest imaginable way, to which one can assign probabilities. Unlike Bell's original realism thesis, it imposes very few constraints on the nature of those facts and especially does not build in any presumption of objectivism. We use the wording from DeBrota and List (2026):

Measurement realism: Any observer's measurement outcomes correspond to facts over which there are well-defined probabilities, where there is no presumption whether the facts are objective or subjective, absolute or relative, coherent or incoherent.

Note especially the "no presumption" clause. The subsequent three theses then rule out that the facts in question could be relative, incoherent, or subjective. We can think of these as capturing Bell's objectivist assumptions. They match three of the theses introduced in the earlier discussion of consciousness. We here paraphrase them slightly.

Non-relationalism: All facts, including those to which measurement outcomes correspond, are of the absolute form "such and such is the case", not of the relative form "such and such is the case, relative to such and such".

Non-fragmentation: The facts that hold at any given world, as well as any assigments of probabilities, are jointly coherent.

One world: Reality is exhausted by one objective world, not multiple subjective worlds.

Since these four subtheses jointly entail realism in Bell's original sense, the Bell inequalities follow from their conjunction, plus locality and measurement independence. To be precise, this presupposes non-solipsism, since the EPR-style scenarios involved in the derivation of Bell's theorem require more than one observer.

Non-solipsism: There can be more than one observer.

We thus have the following result (DeBrota and List 2026):

Heptalemma for quantum mechanics: Locality, measurement independence, measurement realism, non-relationalism, non-fragmentation, one world, and non-solipsism are jointly inconsistent, given the predictions of quantum mechanics. Any six of the seven theses are jointly consistent.

For present purposes, we want to focus on one particular corollary of this result. We obtain it if we group non-relationalism, non-fragmentation, and one world together under the umbrella of objectivism and treat locality and measurement independence as background assumptions.

Corollary of the heptalemma: Measurement realism, non-solipsism, and objectivism are jointly inconsistent, given the predictions of quantum mechanics and given the background assumptions of locality and measurement independence. Any two of the initial three theses – measurement realism, non-solipsism, objectivism – are jointly consistent.

Note that this restatement structurally mirrors the earlier no-go result for consciousness. Although the present result is ultimately a consequence of Bell's theorem, via the heptalemma, it makes Bell's presupposition of non-solipsism explicit and exploits the fact that Bell's realism thesis combines an ontic claim (measurement realism) with an objectivist one (consisting of non-relationalism, non-fragmentation, and one world).

Now, if we take measurement outcomes at face value – treating them as ontic and not merely epistemic, practical, or operational – then measurement realism is very reasonable. Giving it up would entail a very anti-realist or purely operational understanding of quantum mechanics, although some versions of the Copenhagen interpretation appear

to take that route (as discussed in DeBrota and List 2026). Moreover, dropping non-solipsism is hardly plausible. Therefore, if we are committed to locality and measurement independence, we must reject objectivism. Putting this together, we get a remarkably similar dialectic to the one in the case of consciousness, as illustrated in Table 1.

Table 1: No-go results for consciousness and quantum mechanics

	Consciousness	**Quantum mechanics**
Ontic theses	First-personal realism	Measurement realism
Objectivist theses	One world	One world
	Non-fragmentation	Non-fragmentation
	Non-relationalism	Non-relationalism
Non-solipsist theses	Non-solipsism	Non-solipsism
Background theses	N/A	Locality
		Measurement independence

7. Non-objectivist approaches to quantum mechanics

Again, because objectivism is the conjunction of three theses, any one of them can be relaxed on its own. Each of the three routes has received some attention in recent work on quantum mechanics. For a brief review, we will draw on the taxonomy developed in DeBrota and List (2026), to which we refer the reader for more details.

The first route upholds one world and non-fragmentation but gives up non-relationalism. It assumes that the facts to which measurement outcomes correspond are only relative rather than absolute. This is the position taken by relational quantum mechanics (RQM), a prominent approach pioneered by Rovelli (1996, 2025). In RQM, measurement outcomes correspond to facts in a relative rather than absolute sense. Rovelli (2025) notes that this involves a two-fold departure from classical physics: "(a) variables only take value at discrete interactions and (b) the value a variable takes is only relative to the (other) system affected by the interaction." So, in EPR-style scenarios, measurement outcomes are not facts *simpliciter*, but only facts relative to Alice or facts relative to Bob. The total collection of such relativized facts will then be perfectly coherent. RQM gives a very general account of what the relativization parameter could be: it need not be an observer in the conventional sense, but could be any system. So, when we speak of "facts relative to Alice", this means something like "facts relative to Alice understood as the system performing the measurement". Generally, in RQM, facts correspond to interactions between systems; every system-on-system interaction can give rise to corresponding relative facts. Although this is a departure from the objectivist

worldview, it retains the idea that there is a single world and that this world is coherent. It just takes this world to be populated by relative rather than absolute facts.

The second non-objectivist route is the rejection of non-fragmentation. This has also received some attention. Approaches that go in this direction include quantum-logical approaches that offer a non-classical account of the logical structure of possible events. According to them, possible events do not form a Boolean algebra, where the operations of conjunction ("and"), disjunction ("or"), and negation ("not") behave as in classical logic, but they form a lattice in which the classical logical principle of distributivity fails (Wilce 2024).[11] Furthermore, some versions of QBism embrace the "fragmentalist" idea that quantum-mechanical reality might be only locally but not globally coherent. Fuchs suggests the following analogy:

> "An image that might be useful (but certainly flawed) comes from Escher's various paintings of impossible objects. The viewer would initially like to think of them as 2D projections of a three-dimensional object; but he cannot. Now imagine how much worse it would get if we were to have two viewers with two slightly different paintings, each purporting to be a different perspective on 'the' impossible object. Since neither viewer can lift from his own 2D object to a 3D one, there is no way to unify the pictures into a single whole." (Fuchs 2007, quoted in 2023, p. 127–128)

Spelling out such a fragmentalist picture in a formally precise way remains an ongoing project. Abramsky and Brandenburger (2011), for instance, propose the use of mathematical tools from sheaf theory to formalize a fragmentalist interpretation of quantum mechanics.

In some ways the most radical departure from objectivism is the rejection of one world. This is similar to the route discussed above in relation to consciousness, where the notion of a single objective world is replaced by that of distinct subjective worlds: first-personally centred worlds. An advantage of this, compared to the fragmentalist route, is that it allows each of the subjective worlds to be internally consistent and to be composed of facts in an absolute, non-relational sense. The metaphysical cost is the proliferation of postulated worlds.

At first, it may be tempting to think that the Everett interpretation of quantum mechanics makes this move, as it is often called a "many-worlds interpretation" and refers to a "multiverse" (Wallace 2012). However, as noted in DeBrota and List (2026),

[11] Distributivity would require that "p and (q or r)" is equivalent to "(p and q) or (p and r)", and that "p or (q and r)" is equivalent to "(p or q) and (p or r)".

this is not the best way to understand that interpretation, since Everett still treats the wave function as giving rise to a unified reality, which just happens to be branching-tree-like. Chalmers describes this as a "one-big-world interpretation" and argues that it would be a misinterpretation to attribute to Everett the view that "the world literally splits into many separate worlds every time a measurement is made" (1996, p. 347). An interpretation of quantum mechanics that better fits the rejection of one world is the version of QBism described by Mermin (2019). He writes:

> "The fact is that *my* science has a subject (me) as well as an object (my world). *Your* science has a subject (you) as well as an object (your world). *Alice*'s science has a subject (she) as well as an object (her world). I make the same point three times to underline both the plurality of subjects, and the plurality of worlds that each of us constructs on the basis of our own individual experience." (Mermin 2019, p. 5)

Alice's and Bob's records of the facts and probabilities in EPR-style scenarios can then differ from one another without having to fit into a single objective world. There can be distinct subjective worlds and probability assignments for each of Alice and Bob, where these are each internally coherent. In a similar spirit, Fuchs suggests that "[t]hat which surrounds each of us is more truly a pluriverse" (2012, p. 11), and Pienaar (2024) argues that "[a] single space-time is too small for all of Wigner's friends", referring to Wigner's famous puzzle involving two observers. In DeBrota and List (2026), we proposed the labels "Fragmentalist QBism" and "Pluriverse QBism" to distinguish a form of QBism that rejects non-fragmentation from one that rejects one world. Both suggest significant departures from the standard objectivist worldview.

8. Comparing the cases of consciousness and quantum mechanics

We have argued that consciousness and quantum mechanics both challenge the classical objectivist worldview of science, and that they do so in structurally similar ways. In the case of consciousness, if we take first-personal facts at face value and find solipsism unpalatable, then we must deny objectivism. In the case of quantum mechanics, if we interpret measurement outcomes ontically rather than just epistemically, practically, or operationally, and if we wish to uphold locality and measurement independence, then we must also deny objectivism. In each case, the denial can take three distinct forms: a relationalist form, a fragmentalist form, or a many-subjective-worlds form. The first is, in some respects, the least radical, as it retains the idea of a single coherent world, while the last is, in some respects, the most radical, as it abandons the idea of a single world

altogether and takes reality to be irreducibly plural. The middle option – fragmentalism – retains the idea of a single world but abandons the idea of global coherence. Whether this is metaphysically attractive or unattractive is a matter for debate.

Strictly speaking, there is no reason to require that the departure from objectivism must take the same form in the case of consciousness and in the case of quantum mechanics. One could in principle be a fragmentalist about consciousness and a relationalist about quantum mechanics, for instance. Nevertheless, considerations of theoretical parsimony might speak in a favour of a structurally similar treatment of the two cases. Let us now briefly compare the three possible ways of aligning a non-objectivist theory of consciousness with a non-objectivist interpretation of quantum mechanics.

The first route is to embrace relationalism in both cases. In quantum mechanics, as already noted, RQM offers a relational account of facts, and it has some attraction as it retains the idea of a single coherent world, albeit one populated by relational facts. In addition, the research programme of RQM has been pursued for decades now and has raised hopes of offering a viable way of relaxing Bell's realism without overthrowing too much of the spirit of scientific realism. However, with respect to consciousness, as already noted, giving a relational account of first-personal facts would amount to a denial of first-personal realism in the original sense. The fact that I, say Christian, am having Christian's experiences is not merely the fact that relative to Christian, it is true to say "I am having Christian's experiences", but it is the fact that I am having Christian's experiences *simpliciter*. As noted, even the *total* collection of relativized facts would leave out the fact that *I am having Christian's experiences rather than anyone else's*, and so the relativized facts do not genuinely capture the first-personal givenness of my experiences. Recall also the difficulty of spelling out what exactly the relativization parameter of such relationally re-interpreted first-personal facts is supposed to be. If it is supposed to be a "subject", then we must still clarify what a subject is and how subjects fit into reality. RQM seems to offer a clearer solution to the problem of spelling out the relativization parameter, because it implies that the relational facts of quantum mechanics always correspond to relations between two systems.[12] Arguably, then, the relationalist route works better in the case of quantum mechanics than in the case of consciousness. But interestingly, even in the case of quantum mechanics, some physicists, especially in the QBist camp, have expressed misgivings about a relationalist picture, in part because they find it too third-personal. Glick (2021), taking a QBist perspective, notes that relational interpretations "still aim to provide a description of external reality" (p. 9),

[12]However, Brukner (2021, p. 3) has raised some questions about this aspect of RQM, arguing that "not every quantum system can serve as a measuring device or observer" and that "[w]hile every measurement can be seen as an interaction between two systems ..., not every interaction is a measurement".

albeit one where "relational properties replace intrinsic (non-relational) properties for physical systems" (ibid.), and he points out that "[w]hat QBism denies is that models in quantum theory should be viewed as third-personal descriptions of external reality" (p. 4), continuing: "[f]or QBism, all of quantum theory is first-person for the person who happens to be using it" (p. 9).

The foregoing considerations might then lead us to set the relationalist route aside and to prefer one of the other non-objectivist routes. These are indeed favoured by most first-personal realists in the philosophical debate on consciousness (recall the positions discussed by Fine 2005, Merlo 2016, Lipman 2023, and one of us in List 2023, for example), as well as by QBists in quantum mechanics (recall the quotes from Fuchs 2007 and Mermin 2019, for example). The choice, then, is between fragmentalism and a many-subjective-worlds theory.

Fragmentalism retains the idea that first-personal facts and the facts about measurement outcomes hold *simpliciter* and not merely in a relative sense, and it also retains the idea of a single world; the usual identity "reality = the world" is saved. However, it achieves this only at the cost of global incoherence. According to fragmentalism, the collection of facts making up the world constitutes an incoherent patchwork. There are locally coherent subcollections of facts – "fragments" – but these do not coherently fit together. Although tools from non-classical logic might help us to make sense of this, the technical challenges are still formidable, and fragmentalism would require us to abandon or revise many of the entrenched formal tools we rely on in logic, science, and philosophy: from possible worlds to probability functions and probabilistic coherence. For this reason, a fragmentalist theory might be rather costly overall.

A many-subjective-worlds theory, finally, retains the idea that any "world" is coherent, and that worlds are not merely constituted by collections of relational facts. But it does postulate a proliferation of worlds. To make sense of consciousness, we would have to postulate as many first-personally-centred worlds as there are conscious subjects. And to make sense of quantum mechanics, we would have to postulate as many subjective worlds as there are observers, a point explicitly affirmed by Mermin (2019) in the above-quoted passage: my world, your world, Alice's word, and so on.

The theoretical move from objective to subjective worlds may look a bit less unpalatable, however, once we recognize that a many-subjective-worlds approach still allows us to recover a useful notion of an objective world. Objectivity can be defined as invariance under changes in the subject or observer. If we specify a class of admissible such changes, we can define as "objective" any fact that holds across those changes. We can then introduce an equivalence relation on the set of all subjective worlds by deeming any

two subjective worlds to be "third-personally equivalent" or "objectively equivalent" if and only if they coincide with respect to all objective facts. The equivalence classes with respect to this relation can be understood as third-personal or impersonal worlds (List 2023). This way of thinking about objectivity broadly echoes Husserl's idea that objectivity is always rooted in intersubjectivity. Of course, the notion of a third-personal or impersonal world becomes an abstraction under the present proposal, and it is no longer fundamental, but from the perspective of the current approach this is a feature rather than a bug. Worlds, according to the present picture, are no longer fundamentally objective but fundamentally subjective, centred around a subject or observer. According to the many-subjective-worlds approach, this move is precisely what is needed to do justice to the irreducible subjectivity of consciousness or to the puzzles of quantum mechanics. Note that others, too, have proposed a departure from the "one world" picture in order to make sense of both the mind-body problem and quantum mechanics (see in particular Vacariu 2005, 2016).

9. From the book of the world to the library of reality

To provide a more intuitive sense of the structure of each of the different departures from objectivism we have discussed, it is helpful to revisit the famous thought experiment of the "book of the world".[13] The "book of the world" is a gigantic book that contains a complete and coherent record of all the facts that obtain in the world. It exhaustively describes everything that is the case, up to the most fine-grained detail, as it would be seen by a perfectly informed Olympian observer. One might think of it as an omniscient being's full record of what the world is like. A reader of the book could in principle acquire a complete picture of reality.

The objectivist worldview supports the idea of such a book. As Sider (2011) summarizes the view, "[t]here is an objectively correct way to 'write the book of the world'". In practice, of course, it is very hard to write that book, but as an ideal for science, the proponents of the objectivist worldview say, the notion makes sense: the goal of science and philosophy is to write that book. Of course, the objectivist worldview allows that we may encounter all sorts of epistemological and practical difficulties if we try to embark on this project. In reality, the book cannot be written in finite time, and we will at most approximate it. Still, objectivism implies that the "book of the world" is a coherent ideal.

What we have seen is that consciousness and quantum mechanics each challenge that

[13] Here again, we draw on, and borrow some of the wording from, the exposition in List (2025a). We also thank Philip Pettit for suggesting the "library" metaphor used below.

ideal. Let's begin with consciousness. If we cannot adequately accommodate consciousness in our worldview without postulating irreducibly first-personal facts (and we don't think of these as being merely relational), then there cannot exist a single objective book of the world. As it is conventionally characterized, the book of the world is written from a third-personal or impersonal perspective: the "view from nowhere". Its narrator is an Olympian observer describing the world from the outside. A book narrated like this would fail to include any first-personal facts. It would thus be incomplete. So, it would not be *the book of the world*. For example, it would describe Christian and his experiences in third-personal terms, just as it would describe John's and everyone else's experiences in third-personal terms, but it would leave out the first-personal fact that I, say Christian, am having Christian's experiences rather than anyone else's. A complete record of reality, which includes subjective facts, could not be given in the form of a unified objective book. Likewise, consider quantum mechanics. If we accept locality and measurement independence as background assumptions and we retain measurement realism as an ontic assumption, then it will be impossible to provide a coherent and exhaustive enumeration of all physical facts, non-relationally described.

The three departures from objectivism we have discussed each correspond to different ways of revising the ideal of "the book of the world". The relational route corresponds to what is perhaps the least drastic revision. According to relationalism, there could still be a kind of "book of the world", but it would consist of relational facts. So, it would not be a record of everything that is the case *simpliciter*, but rather a record of what is the case relative to what. We might call this "the look-up table of the world": a cross-tabulation of which relational facts hold relative to which relativization parameters. However, there would be a single such look-up table, and it would be coherent. (We noted, however, that this may fail to accommodate genuinely first-personal facts.)

The fragmentalist route would also retain a kind of book of the world, but crucially it would no longer be unified and coherent. It would no longer resemble the traditional genre of non-fiction or fiction, familiar from textbooks or novels with a coherent story, but it would be more akin to the genre of postmodern literature, where a book can be explicitly fragmented and internally contradictory. A classic study of postmodernist fiction describes the difference between modernist and postmodernist fiction as follows:

> "Intractable epistemological uncertainty becomes at a certain point ontological plurality or instability: push epistemological questions far enough and they 'tip over' into ontological questions." (McHale 1987, p. 11)

According to fragmentalism, the book of the world may be compared with a work of

postmodernist fiction. Its internal fragmentation would be a sign of reality's "ontological plurality or instability". However, there is still a single book, albeit of a fragmented kind.

The many-subjective-worlds route, finally, would abandon the notion of a single book of the world altogether. The notion of "the book of the world" would be replaced by that of an entire library, consisting of one subjective book for each subject or observer. We might call it "the library of reality". Each book in that library would not be written by an Olympian third-personal narrator, but it would reflect the perspective of a particular subject or observer. The book of the world written by one of us, which might be titled "The world as I found it" (as Ludwig Wittgenstein 1922 once proposed), would be different from that written by another. My book, say Christian's, would be centred around my (Christian's) perspective, while John's book would be centred around his; similarly for other subjects. And in the case of our two experimenters performing quantum-mechanical measurements, Alice's book would be different from Bob's. Of course, different such books would still have some chapters in common, namely those that cover only objective facts – facts that are invariant under shifts in perspective. But the different books would come apart in their subjective or observer-specific content, which could be substantial. They could not be coherently merged into a single book because such a merged book could not be narrated from a unified perspective. At best, such a hypothetical merger would take us back to the genre of postmodernist literature.

In short, while objectivism identifies "writing the book of the world" (Sider 2011) as an ideal for science and philosophy, the three departures from objectivism revise that ideal: they replace "the book of the world" with a look-up table, a piece of postmodernist fiction, or an entire library, respectively. We may or may not like these alternative ideals, and we may find them less parsimonious than the ideal of a single coherent book, but if we take the challenges from consciousness and quantum mechanics seriously, we may have to come to terms with one of these alternatives.

10. Concluding remarks

We have compared the ways consciousness and quantum mechanics each challenge the classical objectivist worldview of science. At least under certain assumptions, they each conflict with a package of metaphysical theses that jointly make up that worldview. There is very little room for avoiding the conflict. If we set aside the option of embracing solipsism, in the case of consciousness, the only way to avoid the conflict would be to deny first-personal realism altogether. The explanatory burden would then be to show how this can be done without lapsing into some form of illusionism. And in the case of

quantum mechanics, the only way to avoid the conflict would be to reject either locality or measurement independence or measurement realism, and thereby to go against one of three core assumptions to which many scientists are deeply committed.

Consequently, addressing the challenge may require a departure from objectivism. We have mapped out the three main possibilities: the relationalist route, the fragmentalist route, and the many-subjective-worlds route. Although we have not definitively argued for one of these routes, one of us has a preference for the last route (along the lines explained in List 2023). This prompts the methodological question, recognized by Strawson (1966), of what might count as evidence against the conventional thesis of one objective world. If, as Moore (2016, p. 936) writes, "the principal issue" is "whether 'reality' is unified, or whether, on the contrary, it consists of more than one 'world' ", one might wonder what could speak in favour of the many-worlds option. It is unlikely that empirical evidence alone could adjudicate the issue. Rather, the answer, we think, must ultimately come from an inference to the best explanation. The key consideration is which metaphysical worldview works best overall, given the explanatory challenges we are faced with. If accounting for the *explananda* of consciousness and quantum mechanics turns out to conflict with objectivism, and we are unwilling to give up any of the other assumptions leading to that conflict (as detailed in Table 1), then we must relax either non-relationalism, or non-fragmentation, or one world. Which of these routes we should take comes down to the question of which is explanatorily best. It might be that one of the three non-objectivist routes is ultimately more elegant, more illuminating, or better able to accommodate a broad range of phenomena than the others. That would then speak for the route in question.

In conclusion, it is worth briefly returning to phenomenology. The phenomenologists anticipated many of the issues discussed here (again, see French 2023). Husserl wrote:

> "A purely Objective science aims at a theoretical cognizing of Objects, not in respect of such subjectively relative determinations as can be drawn from direct sensuous experience, but rather in respect of strictly and purely Objective determinations: determinations that obtain for everyone and at all times, or in respect of which according to a method that everyone can use, there arise theoretical truths having the character of 'truths in themselves – in contrast to mere subjectively relative truths'." (1969, p. 38, quoted in French 2023, p. 168)

And he famously criticized that picture of science, arguing that it is deeply limited. It focuses only on third-personally quantifiable abstractions and loses sight of the world

of experience of the observer or subject, their "lifeworld", which he thinks should be central to any enquiry. For structural reasons, in particular, conscious experience cannot be adequately accommodated by an objectivist approach. Husserl's critique of the objectivist physicalism of science culminates in his unfinished work *The Crisis of European Sciences and Transcendental Phenomenology* (Husserl 1970). London and Bauer (1939) took inspiration from Husserl's ideas and extended the phenomenological critique to quantum mechanics, as discussed by French (2023). London thought that quantum mechanics requires us to abandon the quest for an objective description of reality and to come to terms with the centrality of the subject:

> "Most of us today feel that this necessary abandonment of a purely objective description of Nature is a profound change in the physical concept of the world. We feel it as a painful limitation of our right to truth and clarity, that our symbols and formulas and the pictures connected with them do not represent an object independent of the observer but only the relation of subject to object. But is this relation not basically the one true reality that we know?" (London, quoted in French 2023, p. 99)

French (2023) goes so far as to suggest that quantum mechanics "re-unites nature and consciousness and London and Bauer's 'little book' [on the interpretation of quantum mechanics] completes Husserl's project" (p. 185) (cf. Pienaar 2025). Be that as it may, we think the no-go results we have discussed here reinforce the phenomenologists' challenge for the objectivist worldview. Even if the problems of consciousness and quantum mechanics are distinct, and we do not simply claim to reduce one mystery to the other, they raise some structurally similar challenges. Comparing those challenges may yield useful lessons on how one might pursue science without objectivism.

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